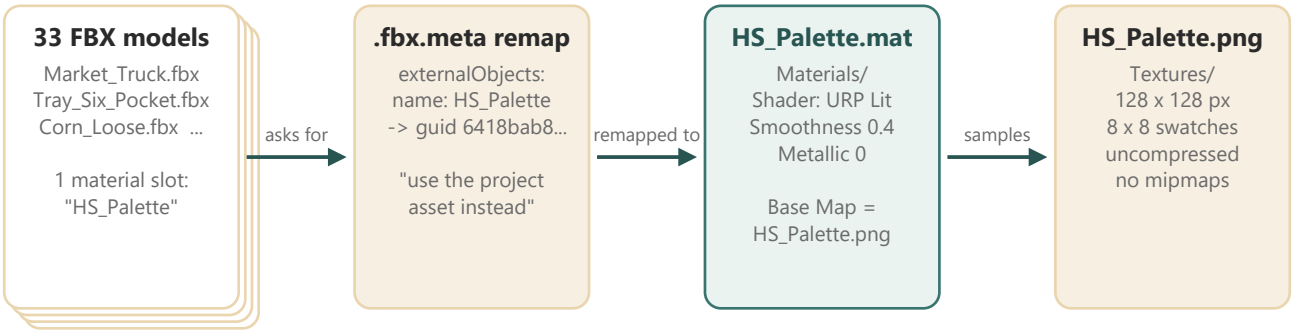
One Material, Every Model

How the Harvest Sorter art pack colours 33 models with a single URP material and a 128 x 128 palette texture, and what that means when you work with it.

The short version

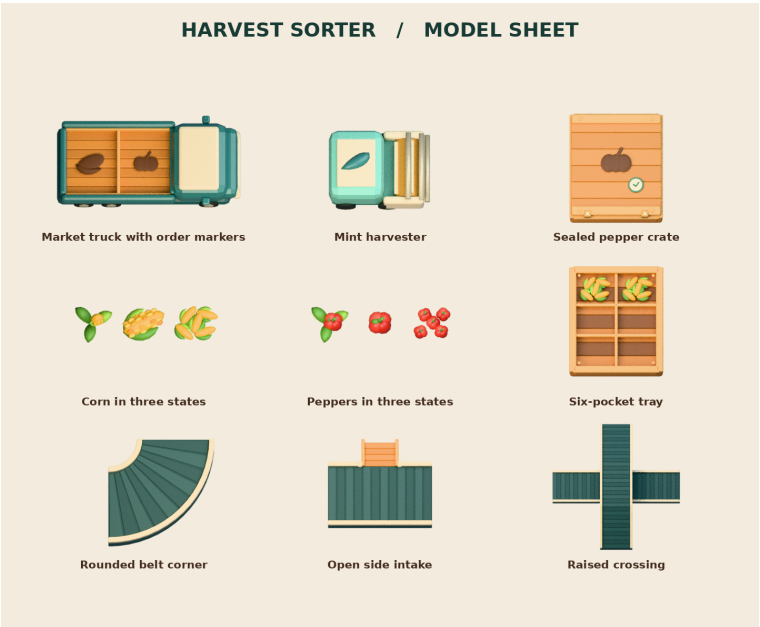
Two things make it work. First, the **models were built that way** in Blender: each one has a single material slot named `HS_Palette`, and its colours come from where its UVs sit on a small grid of colour swatches. Second, each model's **.meta file remaps** that slot to one shared project material, `Materials/HS_Palette.mat`, instead of letting Unity create a separate material inside every FBX.



Every model in the project ends up drawing with the same material and the same texture.

Why do it this way?

- **One place to change the look.** Edit one material or one texture and all 33 models update together.
- **Cheap on mobile.** Every model shares one material and one texture, so Unity can batch them into far fewer draw calls (SRP Batcher / GPU instancing) and there is only one tiny texture in memory.
- **Consistent colours.** Models can only use colours that exist on the palette, so the art stays on-style with no stray one-off materials.



Docs/Art/Model_Sheet.png. The truck, crates, corn, peppers and belts all render with HS_Palette.mat; only their UVs differ.

Part 1: Colour comes from the UVs, not the material

A normal model might have a red material, a wood material and a glass material. These models do not. The material is the same everywhere; the **texture** is a grid of flat colour squares, and each face of a model has its UVs placed inside the square it should be coloured with. The truck's body is teal because its faces point at the teal square.



HS_Palette.png enlarged 4.5x. Numbers match the swatch list below (names and hex values from Art Data/HS_Swatches.json).

Reading a swatch

The texture is 8 x 8 cells of 16 px. A face samples the **centre** of its cell, so the UV coordinates are always at 1/16, 3/16, 5/16 ... (0.0625, 0.1875, 0.3125 ...).

Example, outlined in red: Corn gold

UV = (0.6875, 0.6875)
column = 0.6875 x 8 = **5** (0-based, from the left)
row = (1 - 0.6875) x 8 = **2** (from the top, because UV v=0 is the bottom)
colour = #F8B92C

Cells without a number are unused, spare room for new colours.

Swatch list

1 Ivory enamel	#E9D5AE	**16** Oak rim	#E4A95F	**31** Field soil	#845837
2 Brass	#C59045	**17** Oak slats	#D58A3E	**32** Field soil shade	#6C472F
3 Satin steel	#B5AB8E	**18** Oak slat variation	#C47D38	**33** Field clod	#9E6F47
4 Market petrol teal	#39756F	**19** Oak recess	#794927	**34** Cut stalk gold	#C18F49
5 Truck shadows	#285552	**20** Engraved oak	#663F26	**35** Cut stem top	#E2B775
6 Harvester mint	#73B7A1	**21** Crate seal green	#397350	**36** Warm sand	#D8AF77
7 Harvester trim	#45836E	**22** Corn gold	#F8B92C	**37** Warm sand variation	#CDA674
8 Belt deep teal	#264846	**23** Corn light	#FFD552	**38** Foliage muted green	#729342
9 Belt slat alternate	#2B504D	**24** Corn amber	#E89924	**39** Foliage shadow	#53723A
10 Belt slat groove	#1D3738	**25** Pepper red	#D93B2A	**40** Foliage edge	#8FA650
11 Belt frame	#355955	**26** Pepper light	#EB5137	**41** Barn muted red	#A64F3F
12 Window blue	#35515A	**27** Pepper shadow	#B72E25	**42** Barn plank shade	#934336
13 Window highlight	#688488	**28** Leaf green	#5A923A	**43** Barn slate roof	#69635B
14 Tire	#292E2A	**29** Leaf highlight	#88B345	**44** Covered crop cloth	#6A8E83
15 Honey oak	#B77232	**30** Leaf dark	#417132	**45** Warm white flower	#F1E1B2

Part 2: The .meta remap

When Unity imports an FBX it sees the material slot `HS_Palette`. By default it would build its own material for that slot inside the FBX, which would give 33 separate copies. Instead, every `Models/*.fbx.meta` carries an `externalObjects` entry that says "for the material called HS_Palette, use this project asset":

```
  externalObjects:
  - first:
      type: UnityEngine:Material
      assembly: UnityEngine.CoreModule
      name: HS_Palette
    second: {fileID: 2100000, guid: 6418bab88c9141dfba7219fce8aea16c, type: 2}

# excerpt from Models/Market_Truck.fbx.meta (the same block is in all 33)
```

The `guid` is the ID of `Materials/HS_Palette.mat` (it is in that file's own .meta). Unity links assets by GUID, not by path, so you can move or rename the material and the link still holds. The material itself is a URP Lit material whose **Base Map** is `Textures/HS_Palette.png`.

Seeing it in the Editor

- Select any model in `Assets/Game Assets/Models/`.
- In the Inspector open the **Materials** tab.
- Under **Remapped Materials**, `HS_Palette` points at `HS_Palette.mat`. Changing it here rewrites that model's .meta.

Part 3: Working with it

I want to...	Do this
Change one colour on every model (e.g. a darker teal)	Paint over that swatch in `Textures/HS_Palette.png`. Keep the whole 16 x 16 cell the same colour.
Change the overall look (smoothness, toon shader, rim light...)	Edit `Materials/HS_Palette.mat`, or switch its shader. Every model follows.
Give one object a special look (highlight, ghost, flash)	Assign a different material to that object's MeshRenderer in its prefab or at runtime, or remap it in that one model's Materials tab. Everything else is untouched.
Add a new colour	Paint it into an unused cell, then UV the faces onto it in Blender. Add it to `HS_Swatches.json` so the list stays accurate.
Import a new model from Blender	Give it a single material named `HS_Palette` in Blender. After import, open its Materials tab and set the remap to `HS_Palette.mat` (or press **Search and Remap**). New files do not get the remap automatically.

Rules that keep it working

- **Keep the palette texture uncompressed with mipmaps off.** Compression (ASTC/ETC) averages neighbouring pixels and mipmaps shrink the grid, so colours from adjacent cells bleed into each other. Both are set in `HS_Palette.png.meta`.
- **Keep wrap mode on Clamp.** Swatches on the texture edge would otherwise blend with the opposite edge.
- **Do not resize the palette or move swatches** without re-exporting the models. The UVs are fixed to the current 8 x 8 layout.
- **Do not set the FBX import to 'None' materials** without keeping the remap; the models would fall back to Unity's default grey material.

> **Where the files are**
> `Assets/Game Assets/Models/*.fbx(.meta)` models and their remap
> `Assets/Game Assets/Materials/HS_Palette.mat` the shared material
> `Assets/Game Assets/Textures/HS_Palette.png` the palette
> `Assets/Art Data/HS_Swatches.json` swatch names, hex, UVs